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⇒ Logic

Har composite number ka aik factor pair hota hai.

$\Rightarrow$ Assume number = 36 $\Rightarrow \sqrt{36} = 6$

Factors :

$$1 \times 36 = 36$$

$$2 \times 18 = 36$$

$$3 \times 12 = 36$$

$$4 \times 9 = 36$$

$$6 \times 6 = 36$$

after that repeating again

$$9 \times 4 = 36$$

$$12 \times 3 = 36$$

$$18 \times 2 = 36$$

$$36 \times 1 = 36$$

Yani agar hum ko 6 tk koi factor mil gya to number prime nhi hai, or agar na mila to 6 say aagay b check krnay ka koi faida nhi, koi new information nhi milni.

$\Rightarrow$ Maan lo humary pass 29 hai.

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sub se pehlay iss k factor dhondnay
hein.

$$1 \times 29 = 29$$

iss k mtlb ye prime hai.

$$\text{Ab } \sqrt{29} \approx 5.38$$

Yani algorithm sirf check krega.

2

3

5

check krtay hein.

$$29 \div 2 = 1$$

$$29 \div 3 = 2$$

$$29 \div 5 = 4$$

kisio say divide nhi hua to algorithm bolay
ga that is prime number.

Ab aap soch rhey hon gay k

2, 3, 5 hi kyun

7, 11, 13 ... kyun nhi ?

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Chalo maan laitay hain 29 prime nhi hota, to uss k factor honay chahoye, thay.

$$2 \times ? = 29$$

$$3 \times ? = 29$$

$$5 \times ? = 29$$

factors hi nhi milay, lakin 7 check krtay hain.

$$\text{Ab } 29/7 = 4.14$$

lakin 4.14 to decimal hai, integer hai hi nhi, to factor not hokta.

Conclusion:

$\sqrt{n}$ tk loop chalanay ka yei faida hogya k hamari iterations km hogai and hum ko uss k square-root say aagay janay ki need hi nhi.